

Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Embedded Systems [ECM_3122]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Differentiate between i) Microprocessor and Microcontroller. ii) RISC and CISC. (4)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Discuss the various hardware components in ARM based Embedded device with a neat diagram. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Enlist any three commonly used Assembler directives in ARM along with their description. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Highlight the significance, features of PC relative addressing mode with suitable syntax or examples. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Give the ARM data transfer instruction format indicating the function of each field. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
C) Differentiate the features of N flag and V flag with suitable examples. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Mention the instructions used in Processing of Arithmetic instructions. Discuss the procedure to perform 64 bit addition. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Discuss the significance of Q-notation. Convert $\sqrt{3}/2$ into a Q15 format. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Illustrate and explain Pipelining architecture in ARM. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Compare the looping statements used in ARM with examples. (4)

A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) Extend the significance of Parameters in Subroutine by providing the example of Passing through registers. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Discuss the fsalient features of ARM APCS (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) What is an Exception? Elaborate on the significance of Vector table. (4)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) List the operational quality attributes and define any two of them. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

C) Illustrate and explain domain specific Embedded systems. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	6
3	3	CO 3	20
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	8	27(54%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)